

Executive Summary: E-Commerce Churn Intelligence

End-to-End Data Pipeline, RFM Segmentation, & Predictive Churn Modeling

Author: Huzaifa Rizwan **Role:** Data Analyst Intern **Date:** September 2026 **Stack:** Python, PostgreSQL, Scikit-Learn, Streamlit

1. Project Overview & Business Objective

In modern e-commerce, mitigating customer churn and optimizing Customer Lifetime Value (LTV) are critical pillars for sustainable growth and combating rising Customer Acquisition Costs (CAC). This project establishes a complete, production-ready data analytics and machine learning pipeline using transactional records from a UK-based online retail provider.

The core objective was to transform raw transactional logs into a normalized relational database, derive behavioural insights via **RFM (Recency, Frequency, Monetary)** segmentation, train a robust classification model to predict 30-day customer churn risk, and deploy an interactive executive dashboard (`Streamlit`) for real-time stakeholder decision-making.

2. Key Portfolio Performance Metrics

4,339	66.6%	33.4%	99.1%
ACTIVE CUSTOMER ACCOUNTS	ACTIVE RETENTION RATE	AT-RISK CHURN RATE	MODEL ACCURACY (F1: 0.99)

3. Pipeline Architecture & Execution Summary

- Phase 1 — Data Wrangling & PostgreSQL Normalization:** Ingested 541,909 raw transaction records, resolved ANSI/UTF-8 character encodings, removed 135,080 guest checkouts missing Customer IDs, and filtered out 10,624 return/cancellation anomalies. Structured data into normalized relational tables (`customers`, `products`, `transactions`).
- Phase 2 — Advanced SQL & RFM Metric Extraction:** Executed advanced SQL aggregation scripts to calculate Recency (days since last purchase relative to baseline Dec 10, 2011), Frequency (unique order count), and Monetary (total revenue spend) for every individual customer account.
- Phase 3 — Machine Learning Classification:** Engineered a binary target variable (`is_churned` based on a 90-day inactivity threshold). Trained and evaluated a scikit-learn classification model achieving 99.1% predictive accuracy, successfully serialized via `joblib` (`churn_model.pkl` & `scaler.pkl`).

- **Phase 4 — Interactive Web Application (`Streamlit`):** Built a responsive web app featuring executive KPIs, country filters, dynamic top-tier customer leaderboards, and automated real-time churn risk probability scoring.

4. Top High-Value Customer Accounts (VIP Segment)

The following top-tier accounts represent high monetary value and frequent purchasing behavior, forming the core revenue bedrock of the business:

Customer ID	Country	Recency (Days)	Frequency (Orders)	Monetary Spend (\$)
14646	Netherlands	1	74	\$280,206.02
18102	United Kingdom	0	60	\$259,657.30
17450	United Kingdom	8	46	\$194,550.79
16446	United Kingdom	0	2	\$168,472.50

5. Strategic Recommendations & Action Plan

Key Strategic Takeaway: Rather than allocating budget to broad, generalized acquisition campaigns, the business should deploy targeted, automated retention workflows driven by our ML churn probabilities.

- **VIP Retention Campaigns:** Provide white-glove support, exclusive early access, and loyalty perks to top monetary accounts (e.g., Customer 14646 and 18102) to maintain engagement.
- **Proactive Churn Interventions:** Automatically trigger personalized email win-back offers or discount incentives for customers flagged by our model as approaching the 90-day inactivity threshold.
- **Operational Deployment:** Utilize the deployed Streamlit dashboard for weekly stakeholder reviews to monitor portfolio churn risk across international markets.